Extended Abstract

Mirza Nezam de Gaffary: The Loneliest Mathematician

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Figure 1: Mirza Nezam de Gaffary

This article presents Mirza Nezam de Gaffary, known as Mohandes al-Mamalek, as a singular figure in the history of mathematics in Iran: a mathematician trained at the École Polytechnique in France whose work cannot be properly understood within the usual framework of Qajar-era mathematics. The paper argues that although he lived in that period, he should not simply be classed as a "Qajar mathematician," because the mathematical world to which he belonged intellectually was far closer to nineteenth-century European mathematics than to the

mathematical culture then prevailing in Iran. In that sense, his isolation was double: he was alone in his own lifetime, and his mathematics remained alone after his death, largely unread and unexamined.

A central contribution of the article is archival and documentary. It builds on recently recovered manuscripts and scattered historical sources to offer, for the first time, a broad account of Mirza Nezam's life together with a survey of his surviving mathematical writings. Earlier biographical notices had mentioned him as an important mathematician and had listed some of his works, but they had not examined the mathematical content of those works. This paper therefore tries to do two things at once: reconstruct the historical figure and begin an honest overview of his mathematical production.

The paper places Mirza Nezam's life within a carefully constructed historical timeline. Born in Kashan on 25 August 1844, he belonged to the Ghaffari family and was sent to France for study while still very young, in the late 1850s. In an Iran where even modern schooling was only just beginning, this already marked him as exceptional. He succeeded brilliantly in France, publishing his first mathematical article in French in 1864, even before entering the École Polytechnique in 1865 as the top incoming student. He later studied at the École des Mines and returned to Iran in 1867. The article emphasizes how extraordinary this trajectory was: a young man who had reportedly left Iran with little formal education returned with deep exposure to a living mathematical culture at the highest European level.

The article then traces the striking contrast between Mirza Nezam's mathematical gifts and the roles he was forced to play after returning to Iran. Instead of finding a stable place as a mathematician, he moved through a series of governmental and courtly roles: translator, tutor to Naser al-Din Shah, then to the crown prince Mozaffar al-Din Mirza, road engineer, state official, and occasional minister. He helped translate the Reuter concession, worked on mining and road projects, accompanied royal journeys, and was later among the signatories of the D'Arcy concession in 1901. The paper insists, however, that these public roles should not be mistaken for a life of meaningful scientific influence. In most of them, Mirza Nezam appears as a technically educated man used by the state rather than as a scholar allowed to develop his discipline.

The article gives this tension a human and historical depth. It portrays Mirza Nezam as a man whose life was profoundly shaped by dependence on the Qajar state and its unstable structures of favour, dismissal, and ceremonial office. He was at times close to power, yet never truly established within a lasting scientific institution. Even when he held impressive titles or ministerial posts, these often proved politically shallow or administratively ineffective. The paper thus resists heroic biography. It shows a gifted mathematician whose environment repeatedly diverted him into functions unrelated to the mathematical future he might otherwise have had.

One of the most memorable parts of the article is the comparison with Henri Poincaré. Both entered the École Polytechnique as top students; both published early mathematical work in the same journal, *Nouvelles Annales de Mathématiques*; both went on to engineering schools and later participated in major infrastructure projects. Yet the comparison is deliberately tragic. Poincaré went on to become one of the great mathematicians of modern history. Mirza Nezam, by contrast, returned to a country whose mathematics lagged centuries behind contemporary Europe. The comparison is not meant to inflate Mirza Nezam into "the Persian Poincaré," but to make vivid the scale of the rupture between his training and his context.

The mathematical heart of the article lies in its presentation of Mirza Nezam's large body of surviving manuscripts. These include works on the calculus of variations, algebra, differential and integral calculus, trigonometry, analytic geometry, mechanics, projective and higher geometry, determinants, descriptive geometry, and plane geometry. The paper stresses both their number and their range. Taken together, they cover much of what today would span at least the first years of undergraduate mathematical study and, in places, go well beyond that. The article also records the astonishing speed with which some of these works were written, often over short periods of forced inactivity, illness, or administrative interruption.

Particular attention is given to Mirza Nezam's earliest substantial mathematical work, his 1873 French manuscript on the calculus of variations. The article places this in the wider European development of the subject, noting that even in Europe there were then relatively few pedagogical texts on the calculus of variations. Mirza Nezam's effort is therefore remarkable not only by Iranian standards but also as an educational attempt within an advanced area of nineteenth-century mathematics. Yet the paper also makes clear the historical impossibility of his position: when he wrote such a text, mathematical symbolism itself was only just beginning to secure a foothold in Iran through the much more elementary works of figures such as Abd al-Ghaffar Najm al-Dowleh.

The most original theme of the paper is Mirza Nezam's attempt to create a Persianized mathematical notation. He did not merely translate mathematical terminology into Persian. He tried to redesign the written form of mathematics itself so that it would better fit Persian script and right-to-left writing. He insisted on writing numbers from right to left and developed a system of algebraic symbols derived from Persian letters. The article reproduces and discusses this notation at length, including his arguments for why such a reform was, in his view, more natural for Persian readers. This project stands at the center of the paper's title claim: Mirza Nezam was "the loneliest mathematician" because he created a mathematical language that only he truly spoke.

The article's treatment of this symbolic project is sympathetic but unsentimental. It tries seriously to understand his decision, and even to test whether such a Persian mathematical writing system might conceivably have succeeded. In the end, however, the paper concludes that the project was structurally doomed. The problem was not simply whether numbers should be written from left to right or right to left. The deeper problem was that Mirza Nezam's notation required the creation of a new symbolic alphabet whose successful use depended heavily on handwriting and on a shared community of readers and writers. He had no such community. Unlike the milder symbolic adaptations introduced by Najm al-Dowleh, Mirza Nezam's system cut him off both from European mathematics and from the emerging symbolic mathematics of Iran itself.

This leads to one of the paper's largest historical arguments. Najm al-Dowleh and Mirza Nezam were near contemporaries, and both knew that algebra was, in a sense, "another language." But Najm al-Dowleh chose a form of mathematical writing that connected Iranian learners to international mathematics, whereas Mirza Nezam created a language that remained private. The result was not merely that his notation failed to spread; it also ensured that his books would become inaccessible even within Iran. His work thus represents not a successful founding moment in Iranian mathematical modernity, but an extraordinary dead end—brilliant, creative, and historically revealing, yet ultimately isolated.

The article is equally interested in the content of his books. It argues that many of them de-

serve serious study in their own right, not merely as curiosities. In his algebra, for example, Mirza Nezam moves beyond routine equation solving toward a more structural and functional understanding. In trigonometry, he defines trigonometric functions on the circle rather than beginning only from right triangles. In calculus, he translates Duhamel's infinitesimal approach. In mechanics, geometry, and determinants, he cites and engages with major European mathematicians, often describing them as his contemporaries or former teachers. The paper gives the reader a sense that Mirza Nezam did not merely once study abroad and then stagnate. He remained intellectually connected, as far as circumstances allowed, to ongoing mathematical developments in Europe.

The article also dwells on the materiality of the manuscripts. Many are beautifully written and illustrated, with precise diagrams and careful dating of composition. Marginal notes added years later show that he lived with these books over long periods. The manuscripts were preserved in family hands and only much later deposited in the National Library, without which their scale and significance might have remained unknown. This recovery itself becomes part of the story: the article is not just about a forgotten mathematician, but about the late rediscovery of a body of work that had almost disappeared from intellectual history.

In its concluding assessment, the paper insists that Mirza Nezam should not be measured against the mathematical level of his Iranian contemporaries. Such a comparison is misleadingly easy. His mathematics belongs to a different scale of reference. Yet precisely because no institutional or pedagogical bridge existed to sustain that level in Iran, his work ended with him. The article therefore calls him the loneliest mathematician in modern Iranian history, and perhaps in modern mathematics more broadly: a man educated among active creators of mathematics, who returned to a country unable to receive what he had become, and who then wrote a large mathematical world that no one around him could truly enter.

Overall, the paper makes an important contribution to the history of mathematics in Iran. It combines biography, manuscript recovery, intellectual history, and analysis of notation to show that Mirza Nezam de Gaffary was neither merely a court engineer nor simply an eccentric reformer. He was a mathematically serious figure whose life reveals the depth of the gap between nineteenth-century European mathematics and the institutions of learning available in Iran. His failure was not a failure of talent, but of historical circumstance—and of a symbolic dream that was too solitary to survive.